

Appendix I: From a Plane Wave to the Steering Vector

why $a_n = e^{j\psi_n}$

BLE AoA notes — appendix to Step 3 (uncalibrated MUSIC)

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For readers who have not met wave propagation in vector form. Everything below is the definition of a travelling wave, followed by a dot product.

1 A wave travelling along one axis

Start with a signal that oscillates at frequency f and moves along the z axis:

$$s(z, t) = \cos(2\pi ft - kz), \quad k = \frac{2\pi}{\lambda}. \quad (1)$$

k is the *wavenumber*: how much the phase changes per metre of distance, just as $2\pi f$ is how much it changes per second of time.

Why does (1) travel toward $+z$? Follow one crest. A crest is a point where the argument of the cosine is constant, say zero:

$$2\pi ft - kz = 0 \implies z = \frac{2\pi f}{k} t = f\lambda t = ct.$$

As t increases, the crest's position z increases: the wave moves toward $+z$ at speed $c = f\lambda$. With a $+kz$ sign instead, the crest would satisfy $z = -ct$ and move toward $-z$. So

a *minus* sign in front of the spatial term means propagation in the *positive* direction of that coordinate.

A second reading of (1) is more useful for arrays. Freeze time at $t = 0$ and compare two points, z_1 and $z_2 > z_1$. Their phases are $-kz_1$ and $-kz_2$; the point farther along the propagation direction has the *smaller* (more negative) phase. The wave reaches z_2 later, so at any instant z_2 shows an earlier part of the waveform — it *lags*. Conversely, a point the wave reaches first *leads*.

2 A wave travelling in an arbitrary direction

Let the direction of travel be the unit vector $\hat{\mathbf{k}}$. Nothing in the physics prefers the z axis, so the only change is that “distance along the axis” becomes “distance along $\hat{\mathbf{k}}$ ”. For a point at position

$$\mathbf{r} = x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}},$$

its distance along the propagation direction is the projection $\hat{\mathbf{k}} \cdot \mathbf{r}$. Replace z in (1) by that projection:

$$s(\mathbf{r}, t) = \cos(2\pi ft - k\hat{\mathbf{k}} \cdot \mathbf{r}) = \cos(2\pi ft - \mathbf{k} \cdot \mathbf{r}), \quad \mathbf{k} = k\hat{\mathbf{k}} = k_x\hat{\mathbf{x}} + k_y\hat{\mathbf{y}} + k_z\hat{\mathbf{z}}. \quad (2)$$

$\mathbf{k}$ is the *wavevector*: its direction $\hat{\mathbf{k}} = \mathbf{k}/|\mathbf{k}|$ is the direction of travel and its length $|\mathbf{k}| = 2\pi/\lambda$ is the wavenumber. Check: with $\hat{\mathbf{k}} = \hat{\mathbf{z}}$, $\mathbf{k} \cdot \mathbf{r} = kz$ and (2) reduces to (1).

The surfaces of constant phase are the planes $\mathbf{k} \cdot \mathbf{r} = \text{const}$, all perpendicular to $\hat{\mathbf{k}}$ — hence *plane wave*. Every point on one such plane sees the same phase at the same time. The wave is a plane wave in our setting because the tag is at 295 cm and the array is 12 cm across: over the array the spherical wavefront from the tag is flat to within a small fraction of a degree of phase.

3 Phasor form

A cosine at a single frequency is completely described by its amplitude and its phase. Write it as the real part of a complex exponential:

$$\cos(2\pi ft - \mathbf{k} \cdot \mathbf{r}) = \text{Re}\left\{e^{j(2\pi ft - \mathbf{k} \cdot \mathbf{r})}\right\} = \text{Re}\left\{\underbrace{e^{-j\mathbf{k} \cdot \mathbf{r}}}_{\text{spatial phasor}} e^{j2\pi ft}\right\}.$$

The factor $e^{j2\pi ft}$ is the same everywhere; only the spatial phasor differs from point to point. Dropping the common time factor (and, after the mixing of Step 1, it is gone from the sampled data anyway) the wave at $\mathbf{r}$ is represented by

$$\boxed{s(\mathbf{r}) = e^{-j\mathbf{k} \cdot \mathbf{r}}} \quad (3)$$

The sign is inherited from (1): a point farther along $\hat{\mathbf{k}}$ has a more negative phase (it lags), a point behind has a more positive phase (it leads). Two points $\mathbf{r}_1, \mathbf{r}_2$ differ in phase by $-\mathbf{k} \cdot (\mathbf{r}_2 - \mathbf{r}_1)$, which depends only on their *separation* projected on $\hat{\mathbf{k}}$.

4 From propagation direction to arrival direction

The wave travels *from* the tag *to* the array. In AoA we describe the source by the direction we have to look to see it, the unit vector $\hat{\mathbf{s}}$ from the array centre toward the tag. That is the opposite of the direction of travel:

$$\hat{\mathbf{k}} = -\hat{\mathbf{s}}, \quad \mathbf{k} = -\frac{2\pi}{\lambda} \hat{\mathbf{s}}.$$

In the array frame of Step 3 (+ x boresight toward the tags, + z up, + y to the tag's right) a tag at azimuth AZ and elevation EL is in direction

$$\hat{\mathbf{s}} = \cos \text{EL} \cos \text{AZ} \hat{\mathbf{x}} + \cos \text{EL} \sin \text{AZ} \hat{\mathbf{y}} + \sin \text{EL} \hat{\mathbf{z}} = (\cdot, u, v), \quad (4)$$

where $u = \cos \text{EL} \sin \text{AZ}$ and $v = \sin \text{EL}$ are the direction cosines along y and z . Substituting into (3):

$$s(\mathbf{r}) = e^{-j\mathbf{k} \cdot \mathbf{r}} = e^{+j\frac{2\pi}{\lambda} \hat{\mathbf{s}} \cdot \mathbf{r}}. \quad (5)$$

The two minus signs have cancelled. Read physically: a point with $\hat{\mathbf{s}} \cdot \mathbf{r} > 0$ is displaced *toward the tag*; the wave reaches it first; it leads; its phase is positive. That is the whole content of the steering vector.

5 The steering vector of the array

The sixteen patches lie in the plane $x = 0$ at

$$\mathbf{r}_n = 0 \hat{\mathbf{x}} + y_n d \hat{\mathbf{y}} + z_n d \hat{\mathbf{z}}, \quad y_n, z_n \in \{-1.5, -0.5, +0.5, +1.5\}, \quad d = 40 \text{ mm}.$$

The x -component of $\hat{\mathbf{s}}$ plays no part because every element has $x = 0$ — the array is blind to the boresight component and only senses (u, v) . Putting $\mathbf{r}_n$ into (5):

$$s(\mathbf{r}_n) = e^{j \frac{2\pi}{\lambda} (u y_n d + v z_n d)} = e^{j\psi_n}, \quad \boxed{\psi_n = 2\pi \frac{d}{\lambda} (u y_n + v z_n)} \quad (6)$$

This is the phase of the wave at antenna n , relative to the array centre, for a source at (AZ, EL). Collect the sixteen values into a vector:

$$\mathbf{a}(\text{AZ}, \text{EL}) = [e^{j\psi_1}, e^{j\psi_2}, \dots, e^{j\psi_{16}}]^T, \quad a_n = e^{j\psi_n}.$$

$\mathbf{a}$ is the **steering vector**. It answers the question “if a plane wave arrived from (AZ, EL), what pattern of phases would an ideal array show?” — and MUSIC works by asking, for each candidate direction, how well that ideal pattern fits the measured one.

Three remarks tie this to the rest of the pipeline.

Why the sign is $+j\psi_n$. Because ψ_n was defined from the *arrival* direction $\hat{\mathbf{s}}$, not the propagation direction $\hat{\mathbf{k}}$. Had the steering vector been written directly in terms of $\mathbf{k}$ it would read $e^{-j\mathbf{k} \cdot \mathbf{r}_n}$; the two forms are identical. An antenna displaced toward the source ($u y_n + v z_n > 0$) leads, and a lead is a positive phase. With $d/\lambda = 0.32$ one pitch of displacement along $\hat{\mathbf{s}}$ is $2\pi \cdot 0.32 = 115.2^\circ$.

Referencing to ANT1. The snapshots of Step 2 have ANT1’s phase subtracted, so the steering vector is used in the same form, $a_n \leftarrow a_n \bar{a}_1$, i.e. $\psi_n \leftarrow \psi_n - \psi_1$. Only phase *differences* between elements carry direction information; the absolute phase at the array centre was ϕ_0 in Step 2 and was discarded.

The sign factors s_y, s_z . Equation (6) is the physics. The capture chain may hand us the complex conjugate of every sample (Step 1 showed the mixer convention decides whether a positive frequency offset rotates the I/Q phasor one way or the other, and the same convention governs spatial phase). Conjugating every $e^{j\psi_n}$ is the same as replacing (y_n, z_n) by $(-y_n, -z_n)$, so Step 3 writes

$$\psi_n = 2\pi \frac{d}{\lambda} (s_y u y_n + s_z v z_n), \quad s_y, s_z \in \{+1, -1\},$$

and determines the signs from tags of known position. The result $(s_y, s_z) = (-1, -1)$ on every dataset from this board means the recorded I/Q is the conjugate of (5); it does not mean the array is upside down, which would give a mixed result such as $(+1, -1)$.

6 Worked numbers

Tag at pt46: AZ = 0° , EL = -22.1° , so $u = 0$, $v = \sin(-22.1^\circ) = -0.376$. From (6) with $s_y = s_z = -1$:

$$\psi_n = 115.2^\circ \times (-1)(-0.376) z_n = 43.4^\circ z_n.$$

z_n	+1.5	+0.5	-0.5	-1.5
ψ_n	+65.1°	+21.7°	-21.7°	-65.1°
$\psi_n - \psi_1$ (ref. ANT1, $z_1 = +1.5$)	0	-43.4°	-86.8°	-130.2°

Because $u = 0$ the phase is the same in every column and steps by -43.4° per row going down — the “ideal $\angle a$ ” row of the comparison table in Step 3, where the measured pattern was found to follow it with a shallower slope. That shortfall, not the plane-wave model, is what the calibration steps address.